Example 4.61 By the principle of mathematical induction, prove that, for all integers $n \geq 1$,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Solution:
Let,

$$P(n) := 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Substituting the value of $n = 1$, in the statement we get, $P(1) = \frac{1(1+1)}{2} = 1$. Hence, $P(1)$ is true.

Let us assume that the statement is true for $n = k$. Then

$$P(k) = 1 + 2 + 3 + ... + k = \frac{k(k+1)}{2}.$$

We need to show that $P(k+1)$ is true. Consider,

$$P(k+1) = \underbrace{1 + 2 + 3 + \cdots + k} + (k+1) = \frac{k(k+1)}{2} + (k+1).$$

That is,

$$P(k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}.$$

This implies, $P(k+1)$ is true. The validity of $P(k+1)$ follows from that of $P(k)$. Therefore by the principle of mathematical induction, for all integers $n \geq 1$,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Example 4.62 Prove that the sum of first n positive odd numbers is n^2.

Solution:
Let $P(n) = 1 + 3 + 5 + \cdots + (2n-1)$. Therefore $P(1) = 1 = 1^2$ is true.

We assume that $P(k) = 1 + 3 + 5 + ...(2k-1)$ is true for $n = k$. That is $P(k) = k^2$

We need to prove $P(k+1) = (k+1)^2$.

$$\begin{aligned} P(k+1) &= 1 + 3 + 5 + ...(2(k+1) - 1) \\ &= \underbrace{1 + 3 + 5 + 7 + ... + (2k-1)} + 2k + 1 \\ &= P(k) + 2k + 1 \\ &= k^2 + 2k + 1 = (k+1)^2 \end{aligned}$$

This implies, $P(k+1)$ is true. Hence, by the principle of mathematical induction, $P(n)$ is true for all natural numbers.